HERIOT WATT UNIVERSITY

SCHOOL OF MATHEMATICAL AND COMPUTER SCIENCES

Department of Actuarial Mathematics and Statistics

F78PB

PROBABILITY AND STATISTICS B

IN-PERSON EXAMINATION

April Examination and Assessment Diet 2024

Duration: 2 Hours

Answer **ALL** 6 questions

Attached: Distributions formula sheet, & Correlation and Regression formula sheet

Supplied: Blue Statistical Tables & Yellow Statistical Tables

Approved calculators are permitted

Ethical Commitment: By submitting answers to this paper, I declare that these answers are entirely my own work and have not been shared, in part or in whole or in any form, with any other student, or disseminated in any way.

I also confirm that I have reviewed and understand all University regulations, policies and procedures on assessment misconduct, including but not limited to accessing unauthorised materials during this examination or the use of unauthorised electronic devices.

I understand that infringing this statement would represent an academic misconduct offence subject to disciplinary action according to the University Regulations and Policy regarding Academic Misconduct, with possible significant consequences for degree progression and final degree outcome.

1. Suppose a random variable X has a probability density function defined as:

$$f_X(x) = \begin{cases} e^{-2x} + \frac{1}{2}e^{-x} & x > 0 \\ 0 & \text{otherwise} \end{cases} .$$

a) Show that $M_X(\theta)$, the moment generating function of X is given by:

$$M_X(\theta) = \frac{1}{(2-\theta)} + \frac{1}{2(1-\theta)} \quad \text{for} \quad \theta < 1.$$

[4 marks]

b) Use the $M_X(\theta)$ to determine the $E(X)$ and $Var(X)$, the mean and variance of X.

[7 marks]

c) Given that $Y = \frac{1}{2}X$, find $M_Y(\theta)$, the moment generating function of Y.

[4 marks]

[Total 15 marks]

2. Let X_1, X_2, X_3, X_4 and X_5 are independent and identically distributed random sample of size 5 from a distribution with mean λ and variance λ^2. Suppose we wish to estimate an unknown parameter λ from the distribution based on two candidate estimators:

$$\hat{\lambda}_1 = \frac{1}{8}(X_1 + X_2 + X_3 + X_4) + \frac{1}{2}(X_5)$$
$$\hat{\lambda}_2 = \frac{(X_1 + 2X_2 + 3X_3 + 4X_4 + 5X_5)}{5}$$

a) Calculate the bias for each estimator and explain which one is better. Justify your answer.

[5 marks]

b) Determine the mean square error (MSE) of $\hat{\lambda}_2$.

[3 marks]

c) Given that the MSE of $\hat{\lambda}_1$ is $\frac{5\lambda^2}{16}$, explain why $\hat{\lambda}_1$ is a more efficient estimator than $\hat{\lambda}_2$.

[2 marks]

[Total 10 marks]

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3. a) Let $Y_1, Y_2, ..., Y_n$ be an independent and identically distributed random sample from the probability density function (pdf) defined by:

$$f_Y(y) = \begin{cases} (1-\theta) + 2\theta y & 0 < y < 1 \\ 0 & \text{otherwise} \end{cases}$$

a) Derive a method of moments estimator for θ based on the sample mean $\bar{y}$.

[5 marks]

b) Suppose we take 3 independent random samples from an exponential distribution with unknow paramter λ, and obtain the numbers: 8, 6, 3. Derive the maximum likelihood estimate for λ.

[6 marks]

[Total 11 marks]

4. A company produces two types of energy drinks. The following is a summary for the volumes produced daily:

$$\text{Type A: } n_A = 12, \bar{x}_A = 36,300, S_A^2 = 5,000$$
$$\text{Type B: } n_B = 12, \bar{x}_B = 38,100, S_B^2 = 6,100$$

where $n, \bar{x}$ and S^2 represent the sample size, mean and variance, respectively. Assumed that the volumes are normally distributed with equal population variances.

(a) Explain why

$$\frac{(\bar{x}_A - \bar{x}_B) - (\mu_A - \mu_B)}{\sqrt{S_p^2\left(\frac{1}{n_A} + \frac{1}{n_B}\right)}}$$

is a pivotal quantity for $(\bar{x}_A - \bar{x}_B)$, the difference in population means.

$$\textbf{Note: } S_p^2 = \frac{(n_A - 1)\, S_A^2 + (n_B - 1)\, S_B^2}{n_A + n_B - 2}$$

[3 marks]

(b) For the above data, the 95% confidence interval for the differnce in population mean volumes is calculated as (-1551.37, -1425.44). Does this suggest that the mean volumes produced differ significantly between the two energy drinks produced?

[2 marks]

(c) By constructing an appropriate 95% confidence interval, determine if there is evidence that the population variances are different. Why is this important?
Hint: Use the F distribution

[5 marks]

[Total 10 marks]

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5. Let X and Y be two independent random variables representing the weights in grams of a species of male and female birds, respectively. Assume $X \sim N(\mu_X, \sigma_X^2)$ and $Y \sim N(\mu_Y, \sigma_Y^2)$, where μ_X, μ_Y, σ_X^2 and σ_Y^2 are unknown. You are given $n_X = 16$ observations of X and $n_Y = 13$ observations of $Y, \bar{x} = 415.2, \bar{y} = 347.4, S_X^2 = 1356.8$ and $S_Y^2 = 692.2$.

Define a test statistic and a critical region for testing the hypothesis: $H_0 : \mu_X = \mu_Y$ vrs. $H_1 : \mu_X > \mu_Y$, assuming $\sigma_X^2 = \sigma_Y^2$. Test this pair of hypotheses at 5% significance level and give a clear conclusion.

[Total 6 marks]

6. In a certain type of metal test specimen, the normal stress on a specimen (x) is known to be functionally related to the shear resistance (y). The following is a set of summary data on the variables for $n = 12$ observations:

$$\sum x_i = 311.6, \sum y_i = 297.2, \sum x_i^2 = 8134.3, \sum y_i^2 = 7407.8, \sum x_i y_i = 7687.8$$

(a) Estimate the regression line: $\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_i$ and provide a clear interpretation of $\hat{\beta}_1$ within the given context.

[3 marks]

(b) Test the hypothesis that $\beta_1 = 0$ against the alternative that $\beta_1 \neq 0$ at 5% level of significance.

[3 marks]

(c) A residual plot is commonly used to check the adequacy of the fitted model in **6(a)**. List the three model assumptions that are being checked on this plot.

[2 marks]

[Total 8 marks]

[Grand Total: 60 marks]

END OF PAPER

Some Absolutely Continuous Distributions

Name	Notation	p.d.f.	range	m.g.f.	mean	variance
Uniform	$U(a,b)$	$\frac{1}{b-a}$	$a \le x \le b$	$\frac{e^{b\theta}-e^{a\theta}}{(b-a)\theta}$	$\frac{a+b}{2}$	$\frac{(b-a)^2}{12}$
Normal	$N(\mu,\sigma^2)$	$\frac{1}{\sqrt{2\pi\sigma^2}}\exp\left\{-\frac{(x-\mu)^2}{2\sigma^2}\right\}$	$-\infty < x < \infty$	$e^{\mu\theta+\frac{1}{2}\sigma^2\theta^2}$	μ	σ^2
Gamma	$\Gamma(t,\lambda)$	$\frac{\lambda^t}{\Gamma(t)}x^{t-1}e^{-\lambda x}$	$0<x<\infty$, $t,\lambda>0$	$\left(\frac{\lambda}{\lambda-\theta}\right)^t$	$\frac{t}{\lambda}$	$\frac{t}{\lambda^2}$
Exponential	$\text{Exp}(\lambda)$	$\lambda e^{-\lambda x}$	$0<x<\infty$, $\lambda>0$	$\frac{\lambda}{\lambda-\theta}$	$\frac{1}{\lambda}$	$\frac{1}{\lambda^2}$
Chi-square	χ^2_n	$\frac{1}{2^{n/2}\Gamma(n/2)}x^{n/2-1}e^{-x/2}$	$0<x<\infty$, $n>0$ integer	$(1-2\theta)^{-n/2}$	n	$2n$
Beta	$\beta(\alpha_1,\alpha_2)$	$\frac{\Gamma(\alpha_1+\alpha_2)}{\Gamma(\alpha_1)\Gamma(\alpha_2)}x^{\alpha_1-1}(1-x)^{\alpha_2-1}$	$0<x<1$, $\alpha_1,\alpha_2>0$	not useful	$\frac{\alpha_1}{\alpha_1+\alpha_2}$	$\frac{\alpha_1\alpha_2}{(\alpha_1+\alpha_2)^2(\alpha_1+\alpha_2+1)}$
Cauchy	–	$\frac{\lambda}{\pi(\lambda^2+(x-\mu)^2)}$	$-\infty<x<\infty$, $\lambda>0$	doesn't exist but char. fn. is $e^{i\mu\theta-\lambda\lvert\theta\rvert}$	no moments exist	
Multivariate Normal	$N(\mu,\Sigma)$	$\frac{\exp\left\{-\frac{1}{2}(x-\mu)^T\Sigma^{-1}(x-\mu)\right\}}{(2\pi)^{d/2}\det(\Sigma)^{1/2}}$	$x\in\mathbb{R}^d$	$e^{\mu^T\theta+\frac{1}{2}\theta^T\Sigma\theta}$	μ	covariance matrix Σ

Some Discrete Distributions
(All distributions below have integer-valued ranges)

Name	Notation	prob. fn.	range	m.g.f.	mean	variance
Binomial	$\text{Bin}(n,p)$	$\binom{n}{x}p^x(1-p)^{n-x}$	$0\le x\le n$ $0<p<1$	$[(1-p)+pe^\theta]^n$	np	$np(1-p)$
Poisson	$\text{Pois}(\lambda)$	$\frac{e^{-\lambda}\lambda^x}{x!}$	$0\le x<\infty$ $\lambda>0$ real	$e^{\lambda(e^\theta-1)}$	λ	λ
Hyper-geometric	$\text{Hgeo}(n,a,b)$	$\frac{\binom{a}{x}\binom{b}{n-x}}{\binom{a+b}{n}}$	$0\le x\le n$ $a,b>0$ integers	not useful	$\frac{na}{a+b}$	$\frac{nab(a+b-n)}{(a+b)^2(a+b-1)}$
Negative binomial	$\text{Nbin}(k,p)$	$\binom{x-1}{k-1}p^k(1-p)^{x-k}$	$k\le x<\infty$, $0<p<1$	$p^k[e^{-\theta}-(1-p)]^{-k}$	k/p	$k(1-p)/p^2$
Geometric	$\text{Geo}(p)$	$p(1-p)^{x-1}$	$1\le x<\infty$ $0<p<1$	$p[e^{-\theta}-(1-p)]^{-1}$	$1/p$	$(1-p)/p^2$

Multinomial: multivariate extension of binomial

Prob. fn. is given by

$$f(x_1,x_2,\ldots,x_k)=\frac{n!}{x_!!x_2!\ldots x_k!x_{k+1}!}p_1^{x_1}p_2^{x_2}\ldots p_k^{x_k}p_{k+1}^{x_{k+1}},\quad 0\le x_i\le n,\ \ 0<p_i<1\ \ \forall i=1,2,\ldots,n+1$$

where $p_1+p_2+\cdots+p_{k+1}=1$ and $x_1+x_2+\cdots+x_{k+1}=n$.

The m.g.f. is given by

$$[p_1e^{\theta_1}+p_2e^{\theta_2}+\cdots+p_ke^{\theta_k}+p_{k+1}]^n.$$

Moments: $E(X_i)=np_i$, $\text{Var}(X_i)=np_i(1-p_i)$, $\text{cov}(X_i,X_j)=-np_ip_j$, $i,j=1,2,\ldots,n+1$.

Correlation and Regression

Basic sums:

$$S_{xy} = \sum x_i y_i - \frac{1}{n}(\sum x_i)(\sum y_i)$$

$$S_{xx} = \sum x_i^2 - \frac{1}{n}(\sum x_i)^2 \quad \text{and} \quad S_{yy} = \sum y_i^2 - \frac{1}{n}(\sum y_i)^2$$

Correlation coefficient:

$$r = \frac{S_{xy}}{\sqrt{S_{xx} S_{yy}}} \quad \text{and} \quad t = \frac{r\sqrt{n-2}}{\sqrt{1-r^2}} \sim t_{n-2} \quad \text{under } H_0 : \rho = 0$$

Under the assumption that (x,y) has the bivariate normal distribution, we can test $H_0 : \rho = \rho_0$ against $H_1 : \rho \neq \rho_0$ using the approximation:

$$W = \frac{1}{2}\log\frac{1+r}{1-r} \sim N\left(\frac{1}{2}\log\frac{1+\rho}{1-\rho}, \frac{1}{n-3}\right)$$

Regression parameter estimates:

$$\hat{\beta} = \frac{S_{xy}}{S_{xx}} \quad \text{and} \quad \hat{\alpha} = \frac{1}{n}\{\sum y_i - \hat{\beta}\sum x_i\}$$

$$\hat{\sigma}^2 = \frac{1}{n-2}\{S_{yy} - \frac{S_{xy}^2}{S_{xx}}\}$$

Residuals and fitted values:

Fitted values $\hat{y}_i = \hat{\alpha} + \hat{\beta} x_i$ and Residuals $\hat{e}_i = y_i - \hat{y}_i$

Slope coefficient:

$$t = \frac{\hat{\beta} - \beta}{se(\hat{\beta})} \sim t_{n-2} \quad \text{where} \quad se(\hat{\beta}) = \sqrt{\frac{\hat{\sigma}^2}{S_{xx}}}$$

Prediction standard errors:

For <u>expected</u> Y at $X = x_0$: $\hat{\sigma}\sqrt{\frac{1}{n} + \frac{(x_0 - \bar{x})^2}{S_{xx}}}$ and for <u>individual</u> Y: $\hat{\sigma}\sqrt{\frac{1}{n} + \frac{(x_0 - \bar{x})^2}{S_{xx}} + 1}$